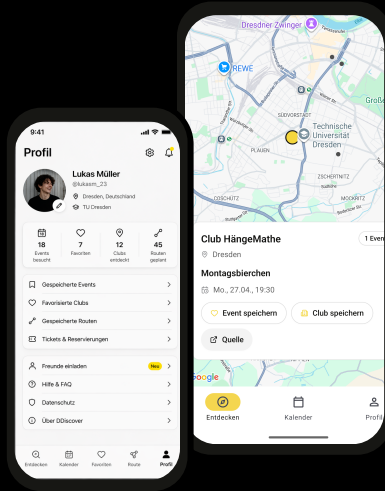


DDiscover

A mobile app for Dresden
student club events



HTW Dresden Mobile Networks APL

Philip Poplutz Gabriel Pechstein

CONTEXT

Problem

Event info is split across club pages, feeds, posts, and word of mouth.

Product answer

One map-first mobile flow for place, time, detail, and route.

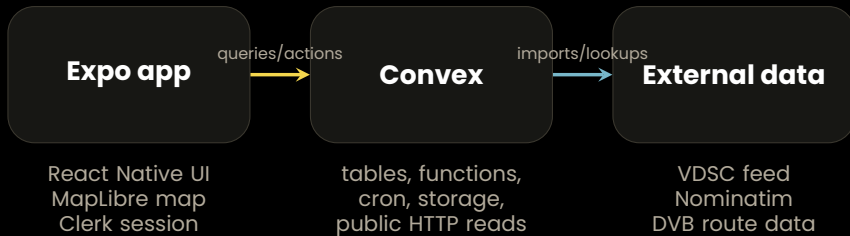
MVP focus

Discover, calendar, details, routes, profile.

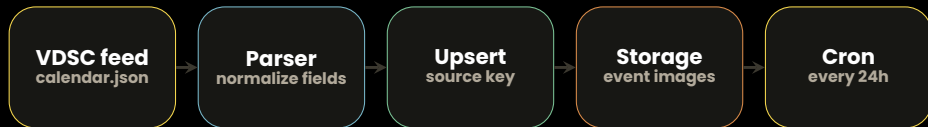
Product question: **what is happening nearby, and how do I get there?**

APP OVERVIEW

SYSTEM SHAPE



IMPORT PIPELINE



Structured VDSC feed first. No browser scraping in the core path.

Images are stored in Convex Storage and exposed through generated URLs.

56+ live VDSC events imported during final testing

LOCATION RELIABILITY

1 Reviewed

Convex row

Manual source wins

2 Fallback

Curated data

Known clubs stay usable

3 Geocoding

Cache first

Nominatim only fills gaps

Lesson: automated geocoding helps, but reviewed coordinates are the better source of truth for a student-facing map MVP.

NATIVE RUNTIME

Runtime issue

The standalone APK exposed behavior that did not appear in local development builds.

Observed

Metro and the dev client loaded data correctly, but the standalone APK failed on Convex data access.

Root causes

Unstable Convex function references in effects, WebSocket behavior in the APK context, and a trailing slash in the CI Convex URL.

Fix

Public app data loads through stable Convex HTTP queries; runtime config strips trailing slashes before building API paths.

TESTING AND CI

Backend tests	19 tests covering parser behavior, imports, geocoding, favorites, and image handling.
Static checks	TypeScript, lint, and Vitest are used as the normal local validation path.
APK builds	GitHub Actions and GitLab CI both prepare Android APK build paths.
Device loop	Build artifact, install on phone, inspect logs, and repeat for runtime-only issues.

Expo Go was not enough because the app uses native modules and a dev-client workflow. APK builds make the final behavior testable on a real Android phone and catch missing public configuration before hand-in.

DIFFICULTIES

Android tooling

ADB, emulator disk, stale
Metro

Auth handoff

Clerk session vs Convex auth
in APK

Data quality

VDSC feed gaps, images,
coordinates

Map interaction

Native markers, drawer
gestures, theme colors

FINAL STATE

- Map-first student club discovery
- Calendar and event detail pages
- Event images stored through Convex
- DVB route planning on event details
- Clerk sign-in + Convex JWT favorites
- Profile settings: language and theme
- Manual APK builds for phone testing

Future plans

- Beta release on Play Store and App Store
- Official VVO API license and registration
- Club outreach for logos and verified content
- Google Places for reliable location data
- Social auth features: friends, sharing, interest counters